

Manual

Cockpit for System Administrators SIS-CSA 4.0_{pe}

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1 Cockpit for System Administrators

Overview

Purpose

The Cockpit for System Administrators allows for an automated notification based on operation statuses of the HYDRA server.

Implementation notes

You use the Cockpit for System Administrators if you wish to receive system-supported warning messages about specific operating statuses of the HYDRA server in order to be able to react in good time.

Integration

The Cockpit for System Administrators provides escalations which can be used within Escalation Management.

Features

System Integration Service (SIS) for the automatic monitoring of:

- Fill levels of the HYDRA database
- Fill levels of the MES server drives

with adjustable limit values. Alert messages for a definable group of employees by e-mail when limit values are exceeded.

2 Escalations for System Administrators

2.1 HYDRA 8 MES Weaver 3.0/3.1

HYDRA 8 MES Weaver 3.0/3.1 provides the following escalations:

Event	Description	Identifiers	Description	Notes
TNR.OFFLINE	Terminal is offline	TNR.TNR	Terminal number	The event is triggered if the time since the terminal's last status message (signal) exceeds the target cycle for status messages.
		TNR.BEZZL	Terminal name	
		TNR.BEZZK	Terminal location	
		TNR.ZYKL:I	Time in seconds that has passed since the last status message (signal).	
		TNR.ZYKL:S	Target cycle for messages (signals) in seconds	
DB.INCREMENT_TOO_LARGE	Fill level of table spaces too large	DB.INC:GR	Increase in percent	The event is triggered if the fill level of single files / table spaces has increased by <DB.INC:GR> within the last few days.
		DB.INC:TG	Period considered	
FILESYS.FILL_LEVEL_EXCEEDED	Fill level of the drive exceeded	FILESYS.FILESYS	File system	
		FILESYS.SUM	File system size	
		FILESYS.SUM:EINH	Unit of file size	
		FILESYS.FREE	Free disk space	
		FILESYS.FREE:EINH	Unit of free disk space	
		FILESYS.FREE:PROZ	Free disk space in percent	
		FILESYS.USED	Used disk space	
		FILESYS.USED:EINH	Unit of used disk space	
		FILESYS.USED:PROZ	Used disk space in percent	
DB.FILL_LEVEL_EXCEEDED	Fill level of database file groups/table spaces exceeded	FILESYS.USEDGR:PROZ	Limit in percent	
		DB.DBSPACE	File group / table space	
		DB.DBSPNUM	Number (Informix)	
		DB.NCHUNKS	Number of chunks (Informix)	
		DB.SUM	Size	
		DB.SUM:EINH	Size unit	
		DB.FREE	Free disk space	
		DB.FREE:EINH	Unit of free disk space	
		DB.FREE:PROZ	Free disk space in percent	
		DB.USED	Used disk space	
		DB.USED:EINH	Unit of used disk space	

Event	Description	Identifiers	Description	Notes
		DB.USED:PROZ	Used disk space in percent	
		DB.USEDGR:PROZ	Limit in percent	

2.2 HYDRA 8 MES Weaver 4.0_{pe}

HYDRA 8 MES Weaver 4.0_{pe} provides the escalations described in section 2.1 HYDRA 8 MES Weaver 3.0/3.1 and the following escalations:

Event	Description	Identifiers	Description	Notes
LOGIN.FAILED	Failed login	LOGIN.USER	User name of login	The event is triggered every time a login fails.
		LOGIN.DATE	Login date	
		LOGIN.TIME	Login time in seconds since midnight.	
		LOGIN.R_IP	Client IP (remote IP)	
		LOGIN.R_HOST	Client host name or IP (remote host name if available)	
		LOGIN.DEV_ID	Device ID of the client (identical to the IP for REST clients, identical to the computer name for SOAP clients)	
		LOGIN.AUTH_TYPE	Type of login attempt: USER_PASSWORD or PIN_LOGIN or SSO (Simple Single Sign On) or ADI (Active Directory Integration)	
		LOGIN.RETURNVALUE	Type of failed login: USER_LOCKED or USER_NOT_FOUND or PASSWORD_WRONG	

3 Configuration Prometheus and Grafana

3.1 Requirements

The system provides the option to output metrics to the monitoring and alert system Prometheus. You can visualize these metrics using the analytics system Grafana.

In combination with other data suppliers providing for example metrics of the operating system and the database management system, this solution can be used to monitor whether the system operates well; in case of problems, you can be informed.

To output metrics you need the **SIS-CSA** license and **Service Pack 15**.



"Prometheus" and "Grafana" are products of third party manufacturers. Both products are neither sold nor supported by MPDV.

For Prometheus MPDV only provides the metrics for query.

For Grafana MPDV provides exemplary dashboards.

3.2 Activation of metrics output

By default, the output of metrics is deactivated. Make the configuration below to enable the metrics output:

```
<InstallDir>\WSP<SystemNo>\config\application.properties
```

```
management.security.enabled=true
```

3.3 Prometheus installation

Find an overview of Prometheus here:

<https://prometheus.io/docs/introduction/overview/>

Install Prometheus. Find instructions on the installation, operation and configuration of the product here:

https://prometheus.io/docs/prometheus/latest/getting_started

Configuration of the Exporter

Extend the Prometheus configuration as follows to have Prometheus query the metrics:

```
<Prometheus folder>\prometheus.yml
```



```
scrape_configs:
- job_name: '<name of configuration>'
  scheme: 'https'
  metrics_path: '/prometheus'
  static_configs:
  - targets: ['<host name of server>:<port>']
```

The port depends on the system queried:

- System 1: 8080
- System 2: 8081
- System 3: 8082
- etc.

Installation of WMI Exporter / Node Exporter

The following Prometheus exporters can be installed to output metrics of the operating system:

- Windows: https://github.com/martinlindhe/wmi_exporter
- LINUX: https://github.com/prometheus/node_exporter

Find notes on the configuration of the adapter here:

https://prometheus.io/docs/introduction/first_steps/

Use at least the following start parameters of the WMI Exporter for the basic functions:

```
--log.format logger:eventlog?name=wmi_exporter
--telemetry.addr :9182
```

If the Microsoft SQL Server is used as database and if this server is on the same host as the system, you can additionally use the following start parameters:

```
--collectors.enabled cpu,cs,logical_disk,net,os,system,mssql,textfile
--collectors.mssql.classes-enabled=accessmethods,bufman,databases,genstats,locks,memmgr,sqlstats
```

MPDV Index dashboard, tab *SQL Server*: Only if the above parameters are set, the panel functions of the dashboard provided by MPDV are active.

Checking the result

After start of Prometheus, enter the following address in a browser:

<http://<name of Prometheus host>:9090>

The relevant web page must then be displayed. In the provided selection of metrics, the metrics of the following naming scheme must be available shortly after: `wsp_services_*` (e.g. `wsp_services_duration_total_seconds_sum`).

3.4 Grafana installation

Find an overview of Grafana here:

<https://grafana.com/>

Install Grafana. Find instructions on the installation, operation and configuration of the product here:

<https://grafana.com/docs/installation/>

Configuration of Prometheus as data source

After start of Grafana, enter the following address in a browser:

<http://<name of Grafana host>:3000>

The relevant web page must then be displayed.

Open the page to configure the data sources. Add Prometheus as data source.

All steps required to visualize system metrics using Grafana are then fulfilled.

Loading example dashboards

MPDV provides some example dashboards. You can download them from the page below and integrate them in your Grafana installation. Use the search function of this page and search for "MPDV".

<https://grafana.com/grafana/dashboards>

To use the MPDV dashboards, install further Grafana panel plug-ins, if required. The requirements of an MPDV panel are listed in the *Dependencies*.

Find the installation instruction for the plug-ins here:

<https://grafana.com/docs/plugins/installation/>

4 Prometheus System Metrics

4.1 Naming

The general rules normally apply for the naming of metrics and labels. The following websites provide a very good overview:

- <https://prometheus.io/docs/practices/naming/>
- https://prometheus.io/docs/concepts/data_model/#metric-names-and-labels

In addition to the general rules, the following naming conventions apply:

- Metrics, which are centrally available for 100 % of all available services, start with **wsp.services.....**
Metrics with this name are exclusively provided by the basic system functions.
- Metrics provided for a subset of services start with **wsp.services.subset.....**
Only use this naming for metrics if their evaluation is useful for all services.
- Metrics only provided for specific services start with **wsp.service.<service name in lower case>.....**
The relevant service documentation describes the relevant labels.

4.2 Metric data types

The following data types are generally available for metrics:

- Summary – is used to total values
(Example: number of processed data records, duration of processing)
For each summary metric, the following parameters are provided:
 - **<name of metric>_sum**
 - **<name of metric>_count**
 - **<name of metric>_max**
- Counter – Metric, which is incremented with each call.
(Example: number of calls, number of errors)
- Timer – is used to total times (in seconds)
(Example: duration of a service)
For each timer metric, the following parameters are provided:
 - **<Name of the metric>_seconds**
- Gauge – current value, not totaled
(Example: current memory usage, current table space usage, alive status)

General note: Only the currently available content of a metric is transmitted at the time Prometheus collects the data.

Some of the data provided, e.g. with gauge, is not "combined" and transferred one by one. Instead, only the last transferred value (which had overwritten the predecessor values) is transferred.

In Prometheus, the relevant Prometheus configuration distinguishes the different systems. The native labels `job` and `instance` are provided to distinguish the data source.

4.3 Standard metrics...

...provided by basic functions

The basic functions provide the following "basic metrics" for services:

Metric name with labels	Type	Value
wsp.services.calls.total <code>{service_domain="<service domain>", service_function="list insert delete update modify copy lock unlock ...", device_id="<client device ID>", user_id="<user ID>", distribution_id="<distribution ID>", processing_result="processed failed" }</code>	Count	Counts service calls
wsp.services.errors.total <code>{service_domain="<service domain>", service_function="list insert delete update modify copy lock unlock ...", device_id="<client device ID>", user_id="<user ID>", distribution_id="<distribution ID>", processing_result="processed failed" }</code>	Count	Counts how often a service is canceled with error
wsp.services.duration.total_seconds <code>{service_domain="<service domain>", service_function="list insert delete update modify copy </code>	Timer	Processing time of service in seconds

<pre>lock unlock ...", device_id="<client device ID>", user_id="<user ID>", distribution_id="<distribution ID>", processing_result="processed failed" }</pre>		
<pre>wsp.services.returned_records.total_count/_max/_sum {service_domain="<service domain>", service_function="list insert delete update copy lock unlock ...", device_id="<client device ID>", user_id="<user ID>", distribution_id="<distribution ID>", resultset_id="<result set ID>" processing_result="processed empty failed ...", }</pre>	Summary	Number of data records in the result sets of the highest level that the service returns to the calling process.

... provided by services

The following metrics with a global naming for all services can be provided by specific services:

Metric name with labels	Type	Value
<pre>wsp.services.subset.records.total_count/_max/_sum {service_domain="<service domain>", service_function="list insert delete update copy lock unlock ...", device_id="<client device ID>", user_id="<user ID>", distribution_id="<distribution ID>", processing_result="processed empty failed ...", data_type="operation_logon_period process_value..." record_type="deleted inserted </pre>	Summary	Number of data records

<pre> updated listed remaining ..." } </pre>		
<pre> wsp.services.subset.files.total_count/_ max/_sum {service_domain="<service domain>", service_function="list insert delete update copy lock unlock ...", device_id="<client device ID>", user_id="<user ID>", distribution_id="<distribution ID>", processing_result="processed empty failed ...", data_type="operations ..." file_type="processed remaining ..." } </pre>	Summary	Number of files
<pre> wsp.services.subset.time_period.total_c ount/_max/_sum {service_domain="<service domain>", device_id="<client device ID>", user_id="<user ID>", distribution_id="<distribution ID>", processing_result="processed empty failed ...", period_type="processed remaining ..." } </pre>	Summary	Period of time

For a precise listing of the metrics specifically provided by the services, refer to section "Metrics of specific services".

4.4 Standard labels

service_domain

This label is used to narrow down the analysis, e.g. to a specific service domain. With standard metrics, the service domain is included in a label and not in a metric name. For this reason, you can optionally perform analyses for all services, e.g. to evaluate the number of inserted data records from all services. This label is always directly created from the service name (character string before last dot).

The metrics, which include the service name already in the metric name, usually do not use this label.

In combination with the label **service_function**, you can filter by a specific service.

service_function

You can use this label to filter services with similar functions. For example, using this label you can make evaluations of all delete services for the runtime. This label is always directly created from the service name (character string after last dot).

The metrics, which include the service name already in the metric name, usually do not use this label.

In combination with the label **service_domain**, you can filter by a specific service.

device_id

Using this label, you can identify for analyses, which client made the request. For example, this can be helpful if several users work on different devices with the same user name.

user_id

Using this label, you can make analyses for the query of specific users. Optionally, you can also compare the queries of different users.

distribution_id

You use this label to distinguish several parallel services having the same task. You usually use these parallel services to distribute the load.

The content of this label is usually directly derived from the service parameter **distribution.id**.

resultset_id

If a service returns several different result sets to the client, you can use this label to identify for each result set how many data records it includes.

processing_result

You use this label as additional filter to identify processings that have actually processed data.

Valid values:

- **processed**
- **empty**
- **failed**

Specific services can also use other values.

data_type

If metrics with the same name are output for different data in a service, then you can use this label to identify the data the metric refers to.

Example: Process data statistics are generated for each operation logon period. When the metrics for the number of "relevant" data records are output, you can use the label **data_type** to distinguish the following metrics:

```
a) wsp.services.subset.records.total_count/_max/_sum
{...,
  data_type="OperationLogonPeriods"
  record_type="processed"
}
```

⇒ Number of processed operation logon periods

```
b) wsp.services.subset.records.total_count/_max/_sum
{...,
  data_type="OperationLogonPeriods"
  record_type="remaining"
}
```

⇒ Number of the operation logon periods that must still be processed (later).

```
c) wsp.services.subset.records.total_count/_max/_sum
{...,
  data_type="ProcessValueStatistics"
  record_type="compressed"
}
```

⇒ Number of compressed process parameter statistics

If a service includes only metrics of one kind of data, then this label is not needed.

In the context of other metrics, you can use this label for similar purposes. For example, with file-based processings you can distinguish the different file types.

record_type

You can use this label to evaluate for all services how many data records were created, deleted, changed, for example.

Valid values:

- **deleted**
- **inserted**
- **updated**

- `listed`

Specific services can also use other values.

file_type

You can use this label to distinguish the different types of files that might be relevant in the context of a service call.

Valid values:

- `processed`
- `remaining`

Specific services can also use other values.

4.5 Metrics of specific services

4.5.1 AnalysisOperationLogonPeriods.importFromManufacturing

The following service-specific metrics are provided by the service `AnalysisOperationLogonPeriods.importFromManufacturing`:

Metric name with labels	Type	Value
<pre>wsp.services.subset.records.total_count/_max/_sum {service_domain="AnalysisOperationLogonPeriods", service_function="importFromManufacturing", distribution_id="", processing_result="processed empty failed", device_id="<client device ID>", user_id="<user ID>", data_type="OperationLogonPeriods", record_type="processed" }</pre>	Summary	Number of processed data records If the service is finished with an error, then the output is 0. In this case, the label <code>processing_result</code> gets the value <code>failed</code>. If no data records are imported, the label <code>processing_result</code> gets the value <code>empty</code>, otherwise <code>processed</code>.

4.5.2 ProcessDataSpecifications.batchInsert

The following service-specific metrics are provided by the service **ProcessDataSpecifications.batchInsert**:

Metric name with labels	Type	Value
<pre>wsp.services.subset.records.total_count/_max/_sum { service_domain="ProcessDataSpecifications", service_function="batchInsert", distribution_id="<content of the (optional) service parameter distribution.id>", processing_result="processed failed", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessDataSpecifications", record_type="inserted" }</pre>	Summary	Number of processed data records If the service is finished with an error, then the output is 0. Otherwise, the number of data records added to the staging table is output. In case of an error, the label processing_result gets the value failed, otherwise processed.

4.5.3 ProcessDataSpecifications.distributeStandard

The following service-specific metrics are provided by the service **ProcessDataSpecifications.distributeStandard**:

Metric name with labels	Type	Value
-------------------------	------	-------

<pre> wsp.services.subset.records.total_count/_ _max/_sum {service_domain="ProcessDataSpecificatio ns", service_function="distributeStandard", distribution_id="<from variable distributionId>", processing_result="processed empty faile d", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessDataSpecifications", record_type="processed" } </pre>	Summary	Number of processed data records If the service is finished with an error, then the output is 0. In this case, the label processing_result gets the value failed. If no data records are processed, the label processing_result gets the value empty, otherwise processed.
<pre> wsp.services.subset.records.total_count/_ _max/_sum {service_domain="ProcessDataSpecificatio ns", service_function="distributeStandard", distribution_id="<from variable distributionId>", processing_result="processed empty faile d", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessDataSpecifications", record_type="remaining" } </pre>	Summary	Number of specification changes waiting to be distributed. If the service is finished with an error, the label processing_result gets the value failed. If no data records were processed, the label processing_result gets the value empty, otherwise processed.

4.5.4 ProcessDataSpecifications.insert

The following service-specific metrics are provided by the service **ProcessDataSpecifications.insert**:

Metric name with labels	Type	Value
-------------------------	------	-------

<pre>wsp.services.subset.records.total_count/_max/_sum { service_domain="ProcessDataSpecifications", service_function="insert", distribution_id="<content of the (optional) service parameter distribution.id>", processing_result="processed failed", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessDataSpecifications", record_type="inserted" }</pre>	Summary	Number of processed data records If the service is finished with an error, then the output is 0, otherwise it is 1. In this case, the label processing_result gets the value failed, otherwise processed.
---	---------	--

4.5.5 ProcessDataSpecifications.aggregatePeriodicStandard

The following service-specific metrics are provided by the service **ProcessDataSpecifications.aggregatePeriodicStandard**:

Metric name with labels	Type	Value
<pre>wsp.services.subset.records.total_count/_max/_sum {service_domain="ProcessDataSpecifications", service_function="aggregatePeriodicStandard", distribution_id="", processing_result="processed empty failed", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessDataSpecifications", record_type="processed" }</pre>	Summary	Number of processed data records If the service is finished with an error, then the output is 0. In this case, the label processing_result gets the value failed. If no data records are processed, the label processing_result gets the value empty, otherwise processed.

<pre>wsp.services.subset.records.total_count/_max/_sum {service_domain="ProcessDataSpecifications", service_function="aggregatePeriodicStandard", distribution_id="", processing_result="processed empty failed", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessDataSpecifications", record_type="compressed" }</pre>	Summary	Number of processed data records If the service is finished with an error, then the output is 0. In this case, the label processing_result gets the value failed. If no data records are processed, the label processing_result gets the value empty, otherwise processed.
<pre>wsp.services.subset.time_period.total.seconds_count/_max/_sum {service_domain="ProcessDataSpecifications", service_function="aggregatePeriodicStandard", distribution_id="", processing_result="processed empty failed", device_id="<client device ID>", user_id="<user ID>", period_type="processed" }</pre>	Summary	Period of time that has been aggregated (in seconds) If the service is finished with an error, then the output is 0. In this case, the label processing_result gets the value failed. If no data records are processed, the label processing_result gets the value empty, otherwise processed.
<pre>wsp.services.subset.time_period.total.seconds_count/_max/_sum {service_domain="ProcessDataSpecifications", service_function="aggregatePeriodicStandard", distribution_id="",</pre>	Summary	Period of time until offset that must still be aggregated (in seconds) If the service is finished with an error, then the output is 0. In this case, the label processing_result gets the

<pre>processing_result="processed empty failed", device_id="<client device ID>", user_id="<user ID>", period_type="remaining" }</pre>		value failed. If no data records are processed, the label processing_result gets the value empty, otherwise processed.
---	--	--

4.5.6 ProcessDataStatistics.aggregateByOperation

The following service-specific metrics are provided by the service **ProcessDataStatistics.aggregateByOperation**:

Metric name with labels	Type	Value
<pre>wsp.services.subset.records.total_count/_max/_sum {service_domain="ProcessDataStatistics", service_function="aggregateByOperation", distribution_id="<from variable distributionId>", processing_result="processed empty failed", device_id="<client device ID>", user_id="<user ID>", data_type="OperationLogonPeriods", record_type="processed" }</pre>	Summary	Number of processed operation logon periods If the service is finished with an error, then the output is 0. In this case, the label processing_result gets the value failed. If no data records are processed, the label processing_result gets the value empty, otherwise processed.

<pre>wsp.services.subset.records.total_count/_max/_sum {service_domain="ProcessDataStatistics", service_function="aggregateByOperation", distribution_id="<from variable distributionId>", processing_result="processed empty failed", device_id="<client device ID>", user_id="<user ID>", data_type="OperationLogonPeriods", record_type="remaining" }</pre>	Summary	Number of operation logon periods that must still be processed (until specified offset). If the service is finished with an error, the label processing_result gets the value failed. If no data records are processed, the label processing_result gets the value empty, otherwise processed.
<pre>wsp.services.subset.records.total_count/_max/_sum {service_domain="ProcessDataStatistics", service_function="aggregateByOperation", distribution_id="<from variable distributionId>", processing_result="processed empty failed", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessValueStatistics" record_type="compressed" }</pre>	Summary	Number of processed data records If the service is finished with an error, then the output is 0. In this case, the label processing_result gets the value failed. If no data records are processed, the label processing_result gets the value empty, otherwise processed.

4.5.7 ProcessValues.aggregateCurveProgression

The following service-specific metrics are provided by the service **ProcessValues.aggregateCurveProgression**:

Metric name with labels	Type	Value
-------------------------	------	-------

<pre> wsp.service.processvalues.aggregatecurve progression.compression.records.total_count/_max/_sum {distribution_id="<content of the (optional) service parameter distribution.id>", processing_result="processed empty failed", device_id="<client device ID>", user_id="<user ID>", compression="<target table of compression>", record_type="processed" } </pre>	Summary	Number of processed raw data records If the service is finished with an error, then the output is 0. In this case, the label processing_result gets the value failed. This parameter is written for each table that must be aggregated (with changing label compression). It is globally written for workplace and process parameters of the pool of orders. If no data records are processed, the label processing_result gets the value empty, otherwise processed.
<pre> wsp.service.processvalues.aggregatecurve progression.compression.records.total_count/_max/_sum {distribution_id="<content of the (optional) service parameter distribution.id>", processing_result="processed empty failed", device_id="<client device ID>", user_id="<user ID>", compression="<target table of compression>", record_type="compressed" } </pre>	Summary	Number of aggregated data records. If the service is finished with an error, then the output is 0. In this case, the label processing_result gets the value failed. This parameter is written for each table that must be aggregated (with changing label compression). It is globally written for workplace and process parameters of the pool of orders.

		If no data records are processed, the label processing_result gets the value empty , otherwise processed .
--	--	---

4.5.8 ProcessValues.batchInsert

The following service-specific metrics are provided by the service **ProcessValues.batchInsert**:

Metric name with labels	Type	Value
<code>wsp.services.subset.records.total_count/_max/_sum</code> <code>{</code> <code> service_domain="ProcessValues",</code> <code> service_function="batchInsert",</code> <code> distribution_id="<content of the</code> <code> (optional) service parameter</code> <code> distribution.id>",</code> <code> processing_result="processed failed",</code> <code> device_id="<client device ID>",</code> <code> user_id="<user ID>",</code> <code> data_type="ProcessValues",</code> <code> record_type="inserted"</code> <code>}</code>	Summary	Number of processed data records If the service is finished with an error, then the output is 0. In case of an error, the label processing_result gets the value failed, otherwise processed.

4.5.9 ProcessValues.distributeStandard

The following service-specific metrics are provided by the service **ProcessValues.distributeStandard**:

Metric name with labels	Type	Value
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<pre>wsp.services.subset.records.total_count/_max/_sum {service_domain="ProcessValues", service_function="distributeStandard", distribution_id="<from variable distributionId>", processing_result="processed empty failed", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessValues", record_type="processed" }</pre>	Summary	Number of processed data records If the service is finished with an error, then the output is 0. In this case, the label processing_result gets the value failed. If no data records are processed, the label processing_result gets the value empty, otherwise processed.
<pre>wsp.services.subset.records.total_count/_max/_sum {service_domain="ProcessValues", service_function="distributeStandard", distribution_id="<from variable distributionId>", processing_result="processed empty failed", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessValues", record_type="remaining" }</pre>	Summary	Number of process data waiting to be distributed. In mode SINGLE_SERIAL, this metric always includes the value 0. If the service is finished with an error, the label processing_result gets the value failed. If no data records were processed, the label processing_result gets the value empty, otherwise processed.

4.5.10 ProcessValues.importFromPccFiles

The following service-specific metrics are provided by the service **ProcessValues.importFromPccFiles**:

Metric name with labels	Type	Value
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<pre>wsp.services.subset.records.total_count/_ _max/_sum {service_domain="ProcessValues", service_function="importFromPccFiles", distribution_id="<from variable distributionId>", processing_result="processed empty faile d", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessValues", record_type="processed" }</pre>	Summary	Number of processed data records If the service is finished with an error, then the output is 0. In this case, the label processing_result gets the value failed. If no data records are processed, the label processing_result gets the value empty, otherwise processed.
<pre>wsp.services.subset.files.total_count/_m ax/_sum {service_domain="ProcessValues", service_function="importFromPccFiles", distribution_id="<from variable distributionId>", device_id="<client device ID>", user_id="<user ID>", file_type="processed" }</pre>	Summary	Number of processed files If the service is finished with an error, then the output is 0. In this case, the label processing_result gets the value failed. If no data records are processed, the label processing_result gets the value empty, otherwise processed.
<pre>wsp.services.subset.files.total_count/_m ax/_sum {service_domain="ProcessValues", service_function="importFromPccFiles", distribution_id="<from variable distributionId>", processing_result="processed empty faile d", device_id="<client device ID>", user_id="<user ID>",</pre>	Summary	Number of files waiting to be processed. If the service is finished with an error, the label processing_result gets the value failed. If no data records were processed, the label processing_result gets the value empty, otherwise

<code>file_type="remaining"</code> <code>}</code>		<code>processed.</code>
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4.5.11 ProcessValues.insert

The following service-specific metrics are provided by the service **ProcessValues.insert**:

Metric name with labels	Type	Value
<code>wsp.services.subset.records.total_count/</code> <code>_max/_sum</code> <code>{service_domain="ProcessValues",</code> <code> service_function="insert",</code> <code> distribution_id="<content of the</code> <code> (optional) service parameter</code> <code> distribution.id>",</code> <code> processing_result="processed failed",</code> <code> device_id="<client device ID>",</code> <code> user_id="<user ID>",</code> <code> data_type="ProcessValues",</code> <code> record_type="inserted"</code> <code>}</code>	Summary	Number of processed data records If the service is finished with an error, then the output is 0, otherwise it is 1. In this case, the label <code>processing_result</code> gets the value <code>failed</code>, otherwise <code>processed</code>.